Automotive Electrical Trainer Kit



Category: **Automotive** Type: **Full SOP** Priority: **High** 4 labs

Covers these lab items: Electrical, ignition, electronic and air-conditioning system; Electronic Ignition System of An Automobile 4-Wheeler; Instruction Kit for Charging and Starting System; Lighting and Wiring System (two-wheeler)

What it is used for

An automotive electrical trainer kit demonstrates vehicle electrical, ignition, starting, charging, electronic, or air-conditioning circuits in a controlled training setup. Correct connections, supervised testing, and safe fault-finding help learners understand system operation without damaging components.

Main parts

- Trainer panel or board supports the electrical components and circuit layout.
- Battery or approved low-voltage power supply provides electrical power.
- Switches and connectors control and link the training circuits.
- Fuses and circuit protection help limit damage from electrical faults.
- Wiring harnesses and terminals represent vehicle electrical connections.
- Ignition, starting, charging, or electronic modules depend on the trainer model.
- Meters, indicator lamps, and test points support measurement and diagnosis.
- Circuit diagrams and instruction materials explain the intended connections.

Before you start

1. Read the trainer manual and identify the circuits included in the specific kit.
2. Ensure the activity is supervised by a qualified instructor or authorised technician.
3. Inspect the power supply, cables, connectors, fuse ratings, and trainer panel for damage.
4. Keep the power switched off while identifying terminals and preparing connections.
5. Review the circuit diagram and verify the required components and test instruments.
6. Remove metal jewellery and prepare suitable PPE and an uncluttered work area.

How to use

1. Place the trainer on a stable, dry workbench and check the circuit diagram.
2. Identify the supply terminals, ground points, switches, loads, and test points.
3. Connect the circuit with power off, following the diagram and specified polarity.
4. Check wiring continuity and confirm that connections are secure before energising the circuit.
5. Set the meter to the correct function and range before making measurements.
6. Switch on the approved supply and operate the circuit using the designated controls.
7. Observe indicator lamps and measure the specified voltage, current, or continuity values.
8. Introduce only instructor-approved faults using the trainer's designated fault switches or method.

9. Switch off the supply before changing connections, replacing fuses, or correcting faults.
10. After testing, restore the original circuit, switch off power, disconnect leads, and record observations.

⚠ Safety

- Use only the voltage and current ratings specified for the trainer kit.
- Never connect or disconnect wires while the circuit is energised unless the trainer procedure explicitly requires it.
- Do not bypass fuses or replace them with a higher rating.
- Keep liquids away from electrical panels, power supplies, and connectors.
- Never test real vehicle high-voltage EV systems using a low-voltage trainer procedure.
- For air-conditioning modules, do not open refrigerant circuits or handle refrigerants without appropriate training and authorisation.

Care & storage

- Switch off and disconnect the power supply before cleaning the trainer.
- Use suitable dry materials to clean panels, connectors, and test leads.
- Inspect cables, probes, fuses, switches, and terminals for wear or damage.
- Store leads and accessories neatly and report defective components before the next session.

Common mistakes to avoid

- Connecting terminals without checking the circuit diagram or polarity.
- Using the wrong meter function or range for a measurement.
- Bypassing a fuse or energising a circuit with loose wiring.
- Changing circuit connections while power is on.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.